GREENLEAF SALES ANALYSIS

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Contents

1. Introduction 3

2. Business Problems 3

3. Objective 3

4. Dataset Description 4

Structure of the Dataset 4

5. Data Cleaning & Preparation 5

Tool Used 5

Data Cleaning Process 5

6. Insights 6

7. Recommendations 6

1. Introduction

GreenLeaf Retail Stores Nigeria is a fictional mid-sized retail chain selling electronics, groceries, household goods, fashion items, and personal care products across seven branches: Port Harcourt, Lagos, Abuja, Kano, Ibadan, Enugu, and Benin City. Over the past two years, GreenLeaf has grown from two locations to seven, and sales volumes have risen steadily each quarter.

Despite this growth, management has noticed that overall profitability has not kept pace with sales volume, and different branches appear to be performing very differently. The company has never had a dedicated analytics function, so all it has to go on is a raw export of transaction-level sales data pulled directly from its point-of-sale systems across all branches.

2. Business Problems

- Some branches seem to be underperforming, but no one can say precisely which ones, or by how much.
- Certain product categories may be driving revenue but not necessarily profit once discounts are factored in.
- Customer payment behaviour (cash vs POS vs bank transfer vs card) and new vs returning customer patterns are not well understood, which makes marketing and operations decisions harder.

3. Objective

- Clean and standardise the raw sales export so it is reliable for analysis.
- Identify top- and bottom-performing branches, product categories, and products.
- Analyse sales trends over time and by payment method and customer type.
- Design a wireframe for a dashboard, then build the dashboard in Excel.
- Deliver clear, data-backed recommendations to GreenLeaf's management.

4. Dataset Description

The dataset contains 1,537 rows and 13 columns of transaction-level sales data covering January to December 2025.

Structure of the Dataset

Column	Description
Transaction ID	Unique identifier for each sales transaction
Date	Date the transaction occurred — formats are inconsistent across rows
Branch	GreenLeaf branch/store location (Port Harcourt, Lagos, Abuja, Kano, Ibadan, Enugu, Benin City)
Region	Geopolitical region the branch belongs to
Product Category	Category of the product sold
Product Name	Specific product sold
Quantity	Number of units sold in the transaction
Unit Price (NGN)	Price per unit, in Naira
Discount (%)	Discount applied to the transaction, if any
Total Sales (NGN)	Total value of the transaction after discount
Payment Method	How the customer paid (Cash, POS, Bank Transfer, Card)
Sales Rep	Staff member who processed the sale
Customer Type	Whether the customer is new or returning

5. Data Cleaning & Preparation

Tool Used

- Ms Excel

Data Cleaning Process

1. Removal of Duplicate Records

Duplicate entries were identified and removed using the “Remove Duplicates function” in Excel. This ensured that each product in the dataset is unique and eliminated redundancy that could distort the analysis.

2. Identification and Treatment of Missing, Invalid, and Unique Values.

The Filter tool was applied across all columns to detect blank cells, invalid entries, and unwanted symbols.

- For the numerical column, missing values were replaced with the **median** of the column to maintain statistical consistency.
- For categorical columns, blank cells, invalid, unique, and unknown values were replaced with **"the mode of the column"** to preserve dataset completeness and avoid bias during analysis.

3. The Date Column

The column was corrected using the: text-to-column function.

4. Standardization of Data Formats

All columns were standardized to appropriate data formats:

- Numerical columns were formatted as numbers.
- Text-based columns were formatted consistently.

5. Cleaning of Text Fields

Text columns were cleaned to improve readability and consistency by:

- Removing extra spaces using the **TRIM** function.
- Removing unwanted symbols and inconsistencies using **Find & Replace.**

- Inconsistent casing was changed to proper casing using **PROPER**.

6. Insights

- Kano recorded the highest branch sales at ₦29.38M, while Abuja recorded the lowest at ₦18.69M.
- Kano recorded the highest total discount at ₦1.07M, while Ibadan recorded the lowest at ₦495.55K. Although Kano had both the highest sales and discount, the dashboard does not establish a causal relationship between discounts and sales.
- Groceries was the highest-performing product category, generating ₦39.34M in sales, while Beauty & Personal Care recorded the lowest among the displayed categories at ₦26.68M.
- October 2025 recorded the highest monthly sales at approximately ₦13.98M. Overall, monthly sales fluctuated throughout the year without a consistent upward or downward trend.
- Cash was the most commonly used payment method at 31%, followed by Card at 25%, Bank Transfer at 23%, and POS at 21%.
- Returning customers accounted for 53% of customers, compared with 47% for new customers, indicating a slightly larger share of returning customers during the period analyzed.

7. Recommendations

- Investigate the factors contributing to monthly sales fluctuations, particularly the decline following the October sales peak.
- Analyze differences in branch performance using transaction volume, average order value, product mix, discounts, and customer type.
- Evaluate discount effectiveness by comparing discount levels with sales, transaction volume, and average order value across branches and product categories.
- Implement a structured customer feedback system to capture complaints and suggestions and analyze the resulting data alongside customer retention and sales performance.